Table 1. Octinoxate: MM/GBSA binding free energy, ΔG (kcal/mol), and ligand residence (%) from 20 ns MD, started from each of three poses in each of three models' UNFORCED co-folds, plus the pocket-guided OpenFold3 lane.

| peptide | pose | Boltz / BoltzGen co-fold | OpenFold3 co-fold | RoseTTAFold3 co-fold | OpenFold3 pocket-guided |
|---|---|---|---|---|---|
| orig | co-fold | −7.24 (29.9%) • | −5.90 (7.4%) ↑ | **−19.51 (99.9%) ↑** | −8.06 (21.1%) ↓ |
| | Vina p1 | −2.81 (0.6%) ↑ | −9.65 (50.2%) ↑ | **−18.71 (89.2%) ↑** | −12.27 (51.0%) • |
| | GNINA pick (p1/p1/p5/p4) | see Vina p1 | see Vina p1 | −11.10 (71.0%) ↑ | *running* |
| s2_esm2_control | co-fold | −16.25 (59.6%) • | −10.21 (33.9%) ↓ | −14.98 (57.2%) ↓ | −16.28 (100.0%) • |
| | Vina p1 | −15.49 (0.0%) ↓ | −17.27 (95.2%) ↓ | −15.86 (78.9%) • | −16.72 (32.4%) ↑ |
| | GNINA pick (p6/p5/p4/p8) | −17.09 (50.3%) • | −6.84 (9.4%) ↓ | −15.41 (73.7%) ↓ | *running* |
| s3_esm2 | co-fold | −15.70 (29.4%) ↑ | −13.76 (84.8%) ↑ | −16.18 (88.2%) ↑ | −12.71 (86.9%) • |
| | Vina p1 | −9.65 (2.4%) ↓ | −14.53 (12.6%) • | **−20.39 (99.4%) •** | −17.12 (41.6%) ↓ |
| | GNINA pick (p7/p8/p3/p6) | −16.64 (0.1%) • | −12.79 (6.9%) ↑ | **−20.79 (99.3%) ↑** | *running* |
| s3_orig | co-fold | −13.37 (48.0%) ↑ | −10.10 (44.8%) ↑ | −12.10 (68.9%) ↑ | −9.78 (43.0%) ↑ |
| | Vina p1 | −9.41 (36.8%) • | −8.92 (25.2%) • | **−19.16 (90.0%) ↓** | −11.50 (58.0%) ↓ |
| | GNINA pick (p9/p6/p5/p9) | −11.51 (37.5%) ↑ | −11.17 (22.9%) ↑ | −8.43 (7.2%) ↑ | *running* |
| shuffle_control | co-fold | −15.13 (75.9%) ↑ | −9.39 (25.7%) ↓ | −11.13 (38.7%) • | — |
| | Vina p1 | −16.15 (57.8%) | −10.89 (23.3%) • | −5.37 (21.8%) ↑ | — |
| | GNINA pick (p7/p8/p7/—) | −8.40 (22.4%) ↑ | −7.78 (12.6%) ↑ | −13.86 (60.3%) ↓ | — |
| shuffle_control_esm0 | co-fold | −14.32 (80.2%) ↓ | −15.62 (92.5%) ↓ | **−21.53 (99.3%) •** | — |
| | Vina p1 | −15.50 (76.1%) ↑ | −12.27 (1.9%) ↓ | −12.47 (44.1%) • | — |
| | GNINA pick (p9/p3/p3/—) | −14.76 (90.5%) • | **−18.19 (98.5%) •** | −12.03 (36.9%) ↑ | — |
| bg33_1 | co-fold | −11.31 (14.5%) ↓ | −14.90 (56.4%) ↓ | −12.43 (68.3%) ↓ | — |
| | Vina p1 | −13.47 (44.7%) ↓ | −13.16 (22.6%) • | −14.02 (60.3%) ↓ | — |
| | GNINA pick (p2/p6/p8/—) | −12.55 (42.5%) ↓ | −12.93 (44.5%) ↑ | −11.78 (62.6%) • | — |
| bg33_2 | co-fold | −8.02 (50.9%) ↑ | −11.58 (70.2%) • | −11.52 (58.7%) ↓ | — |
| | Vina p1 | −11.82 (43.1%) ↑ | −11.55 (14.8%) • | −10.95 (81.5%) • | — |
| | GNINA pick (p4/p1/p8/—) | −10.44 (57.2%) ↓ | see Vina p1 | −12.19 (36.7%) • | — |
| bg33_3 | co-fold | −11.86 (41.0%) ↓ | −10.10 (14.9%) • | −12.44 (1.9%) • | — |
| | Vina p1 | −10.84 (25.7%) ↑ | −14.89 (99.7%) • | −11.01 (0.2%) ↑ | — |
| | GNINA pick (p6/p6/p5/—) | −13.09 (77.6%) ↑ | −14.92 (99.6%) • | −11.47 (0.1%) ↑ | — |
| bg33_4 | co-fold | **−19.66 (100.0%) ↓** | −17.11 (93.0%) ↓ | −16.13 (81.2%) ↑ | — |
| | Vina p1 | −14.81 (59.3%) ↑ | −16.04 (99.2%) • | **−18.59 (97.6%) ↓** | — |
| | GNINA pick (p6/p7/p4/—) | −16.53 (97.2%) • | −16.24 (82.8%) ↑ | **−19.06 (99.6%) •** | — |

Values are ΔG in kcal/mol with residence in parentheses — the share of the whole 20 ns run with the ligand within 10 Å of the peptide. Each peptide carries three sub-rows: the co-fold pose each model produced, Vina p1 (AutoDock rank 1 docked into that same fold), and the GNINA pick (max CNNaffinity over poses ≥1). The pick's pose in each lane is in the row label, in column order; where it is p1 the pick coincides with Vina's rank 1, so no separate leg was ever run and the cell says *see Vina p1* rather than repeating the number. The trailing marker compares the 20 ns ΔG with the ΔG over the first 10 ns: • stable (within 0.5 kcal/mol), ↓ still decreasing (binding strengthening), ↑ still increasing (binding weakening) — a convergence flag on ΔG, not on residence. Bold marks ΔG ≤ −18. Every cell is a finished 20 ns leg; the campaign completed 2026-10-19.

Residence is a centre-of-mass measure, so read it with the fold's shape in mind. It is the share of frames with the ligand's centroid within 10 Å of the peptide's, which penalises an elongated fold: a ligand gripped at one end of a rod is never released and still scores low, because the far half of the peptide drags the centroid away. s3_esm2 co-fold is the case in point — 29.4% residence with 0.15% of frames detached, i.e. essentially permanent atomic contact. Where retention is the question, the release statistics in md_contacts.csv (episode count, longest episode, late share) are the shape-independent measure.

Table 2. Oxybenzone: MM/GBSA binding free energy, ΔG (kcal/mol), and ligand residence (%) from 20 ns MD, started from each of three poses in each of three models' UNFORCED co-folds, plus the pocket-guided OpenFold3 lane.

| peptide | pose | Boltz / BoltzGen co-fold | OpenFold3 co-fold | RoseTTAFold3 co-fold | OpenFold3 pocket-guided |
|---|---|---|---|---|---|
| ox1_esm1 | co-fold | **−26.09 (100.0%) ↓** | −17.95 (100.0%) ↑ | −12.12 (35.2%) ↓ | −10.87 (95.3%) • |
| | Vina p1 | −10.23 (68.6%) ↑ | −10.60 (89.9%) ↑ | −7.26 (27.3%) ↑ | −8.80 (45.2%) ↑ |
| | GNINA pick (p4/p2/p4/p8) | −10.11 (56.2%) ↑ | −12.79 (66.9%) • | −14.92 (79.8%) ↓ | *running* |
| ox1_orig | co-fold | −8.34 (63.8%) ↑ | −3.96 (20.5%) ↑ | −12.27 (89.1%) ↓ | **−18.26 (100.0%) •** |
| | Vina p1 | −9.34 (61.6%) • | −4.93 (25.5%) ↑ | −6.40 (37.5%) ↑ | *running* |
| | GNINA pick (p6/p4/p1/p4) | −4.79 (17.0%) ↑ | −5.84 (33.0%) ↓ | see Vina p1 | *running* |
| ox2_esm1 | co-fold | −12.70 (71.0%) ↑ | −16.68 (93.2%) ↑ | −10.90 (90.4%) ↑ | −9.21 (31.8%) • |
| | Vina p1 | −4.80 (21.2%) ↑ | −16.87 (93.9%) ↓ | −11.47 (79.1%) ↓ | *running* |
| | GNINA pick (p9/p3/p5/p6) | −11.09 (86.2%) ↓ | −10.58 (77.1%) ↑ | −9.27 (62.4%) ↓ | *running* |
| ox2_orig | co-fold | −8.51 (43.9%) ↑ | −7.23 (49.5%) ↑ | −4.95 (19.6%) ↑ | −13.93 (68.7%) ↑ |
| | Vina p1 | −6.71 (44.0%) • | −9.05 (47.2%) ↑ | −4.22 (15.5%) • | *running* |
| | GNINA pick (p9/p2/p7/p5) | −3.50 (17.0%) ↑ | −5.43 (37.1%) ↑ | −4.11 (8.2%) • | *running* |
| ox2_shuffle | co-fold | −5.30 (7.1%) • | −2.98 (14.8%) ↑ | −13.35 (51.0%) • | — |
| | Vina p1 | −5.86 (26.2%) • | −5.91 (27.3%) ↓ | −8.60 (72.1%) ↓ | — |
| | GNINA pick (p4/p1/p2/—) | −2.09 (1.2%) • | see Vina p1 | −9.29 (32.0%) ↓ | — |
| ox2_shuffle_esm0 | co-fold | −12.04 (94.7%) • | −0.92 (1.1%) • | −11.44 (32.6%) • | — |
| | Vina p1 | −12.78 (97.4%) • | −9.44 (60.4%) ↓ | −9.18 (54.8%) ↑ | — |
| | GNINA pick (p9/p6/p1/—) | −11.87 (95.5%) • | −6.84 (56.3%) ↑ | see Vina p1 | — |
| ox3_orig | co-fold | −10.13 (72.0%) ↑ | −2.06 (2.4%) ↓ | −7.41 (45.6%) ↓ | −10.05 (38.6%) ↓ |
| | Vina p1 | −14.33 (97.9%) ↓ | −4.12 (11.8%) ↑ | −4.32 (19.5%) • | *running* |
| | GNINA pick (p6/p4/p8/p1) | −15.44 (100.0%) ↑ | −5.99 (33.2%) • | −4.90 (10.0%) ↑ | see Vina p1 |
| bgox31_2 | co-fold | −10.89 (33.7%) ↑ | −7.45 (65.9%) ↑ | −9.06 (84.6%) ↓ | — |
| | Vina p1 | −2.11 (8.8%) ↑ | −5.85 (10.0%) ↓ | −7.36 (47.9%) ↑ | — |
| | GNINA pick (p6/p1/p3/—) | −8.50 (2.6%) • | see Vina p1 | −9.57 (52.9%) • | — |
| bgox31_3 | co-fold | −15.44 (99.7%) ↑ | −5.82 (15.8%) ↑ | −14.36 (99.2%) • | — |
| | Vina p1 | **−22.23 (100.0%) ↓** | −10.99 (68.5%) ↑ | −13.48 (95.7%) ↑ | — |
| | GNINA pick (p3/p4/p1/—) | −17.52 (89.9%) ↑ | −12.48 (88.9%) • | see Vina p1 | — |
| bgox31_4 | co-fold | −14.05 (100.0%) ↑ | −5.74 (17.9%) ↑ | −11.02 (72.0%) • | — |
| | Vina p1 | −9.20 (36.5%) ↑ | −10.39 (49.9%) ↑ | −15.86 (99.9%) • | — |
| | GNINA pick (p6/p7/p3/—) | −4.31 (4.9%) ↑ | −6.71 (11.3%) ↓ | −13.91 (100.0%) • | — |
| bgox31_5 | co-fold | −16.78 (100.0%) • | −11.55 (70.7%) ↓ | −14.69 (96.4%) ↑ | — |
| | Vina p1 | −12.81 (70.3%) • | −13.72 (93.4%) • | −14.93 (99.8%) • | — |
| | GNINA pick (p2/p7/p8/—) | −14.28 (99.5%) • | −14.84 (99.5%) ↑ | −14.76 (94.9%) • | — |

Values are ΔG in kcal/mol with residence in parentheses — the share of the whole 20 ns run with the ligand within 10 Å of the peptide. Each peptide carries three sub-rows: the co-fold pose each model produced, Vina p1 (AutoDock rank 1 docked into that same fold), and the GNINA pick (max CNNaffinity over poses ≥1). The pick's pose in each lane is in the row label, in column order; where it is p1 the pick coincides with Vina's rank 1, so no separate leg was ever run and the cell says *see Vina p1* rather than repeating the number. The trailing marker compares the 20 ns ΔG with the ΔG over the first 10 ns: • stable (within 0.5 kcal/mol), ↓ still decreasing (binding strengthening), ↑ still increasing (binding weakening) — a convergence flag on ΔG, not on residence. Bold marks ΔG ≤ −18. Every cell is a finished 20 ns leg; the campaign completed 2026-10-19.

Residence is a centre-of-mass measure, so read it with the fold's shape in mind. It is the share of frames with the ligand's centroid within 10 Å of the peptide's, which penalises an elongated fold: a ligand gripped at one end of a rod is never released and still scores low, because the far half of the peptide drags the centroid away. s3_esm2 co-fold is the case in point — 29.4% residence with 0.15% of frames detached, i.e. essentially permanent atomic contact. Where retention is the question, the release statistics in md_contacts.csv (episode count, longest episode, late share) are the shape-independent measure.

Table 3. The strict comparison, static half: one sequence, four co-folding treatments.

| peptide | Boltz forced | Boltz unforced | OF3 unforced | RF3 unforced |
|---|---|---|---|---|
| orig_f12 (oct) | 0.960 / 4.2 Å | **0.495 / 20.9 Å** | 0.680 / 9.4 Å | **0.800 / 5.8 Å** |
| s3_esm2_f4 (oct) | 0.795 / 7.0 Å | 0.360 / 13.6 Å | **0.580 / 7.3 Å** | 0.460 / 9.3 Å |
| s3_orig_f12 (oct) | 0.965 / 3.6 Å | **0.735 / 9.5 Å** | 0.430 / 10.4 Å | 0.720 / 8.5 Å |
| ox1_esm1_f8 | 0.940 / 4.1 Å | **0.765 / 5.9 Å** | 0.580 / 6.1 Å | 0.530 / 9.3 Å |
| ox1_orig_f12 | 0.995 / 1.2 Å | 0.520 / 7.2 Å | **0.810 / 5.7 Å** | 0.550 / 6.1 Å |
| ox2_esm1_f8 | 0.665 / 6.4 Å | 0.610 / 7.6 Å | **0.745 / 5.3 Å** | 0.570 / 6.4 Å |
| ox2_orig_f8 | 0.975 / 2.7 Å | 0.535 / 9.0 Å | **0.655 / 7.5 Å** | 0.595 / 6.2 Å |
| ox3_orig_f8 | 0.995 / 2.1 Å | **0.615 / 5.8 Å** | 0.540 / 10.3 Å | 0.465 / 11.4 Å |
| **mean** | **0.911 / 3.9 Å** | 0.579 / 9.9 Å | 0.628 / 7.8 Å | 0.586 / 7.9 Å |

Each cell is enclosed fraction / ligand-centroid separation. Bold marks the best enclosure among the three *unforced* lanes in that row (Boltz forced is excluded from that contest — it is the treatment under test, not a competitor).

What the strict comparison says, and it is not what Tables 1 and 2 imply. The three unforced lanes are indistinguishable on this measure: mean enclosure 0.579 (Boltz), 0.628 (OF3), 0.586 (RF3), with per-peptide wins splitting 3 / 4 / 1 across eight sequences and every lane's range overlapping the others' almost completely. Forced Boltz is in a different regime from all three — 0.911 enclosed at 3.9 Å, against 0.58–0.63 at 7.8–9.9 Å — and no unforced lane's best cell (0.810) reaches forced Boltz's mean. So the Boltz column's advantage in Tables 1 and 2 is attributable to the contact hints, not to Boltz being the better co-folder: strip the hints and it places the ligand no better than the other two, and on octinoxate's orig_f12 it is the worst of the three by a wide margin (0.495 at 20.9 Å, the furthest starting pose in the whole campaign).